

Team DNA Hackers

Project Phase 3

Team Members:

Janya Kaushal Bhatt

Pariza

Sara Bansal

1 - Mapping Er To Relational Model

1. Mapping of Regular Entity Types and Simple Attributes

- Rule: Each strong entity type (e.g., {Agent}, {Mission}, {Location}) was converted into a base relation.
- Keys: The primary key for each entity (e.g., {AgentID}) was designated as the Primary Key (PK) for the corresponding relation.
- Attributes: All simple, single-valued attributes ({FirstName}, {Country}, {Status}, etc.) were included directly in their respective relations.
- Composite Attributes: We flattened composite attributes, such as the contact information, by including only their simple component fields ({Email}, {Phone}) in the {Agent} relation, based on our decision to treat them as single-valued for the scope of this project.

2. Mapping of Weak Entity Types and Identifying Relationships

- Rule: The {Evidence}, {MissionTask}, and {ArtifactMovement} entities, which cannot exist without their owner entities, required special handling.
- Primary Key Definition: The Primary Key for these new relations is always a composite key formed by:
 1. The Primary Key of the Owner Entity (as an FK).
 2. The Partial Key of the weak entity itself.
- Example: For {MissionTask}, the Primary Key became {MissionID, TaskNo}).

3. Mapping of N:1 and 1:1 Relationship Types

- Rule (1:N): We did not create new relations for 1:N relationships. Instead, we included the Primary Key of the '1'-side entity as a Foreign Key (FK) in the relation on the 'N'-side.
- Example: The {Collects} relationship ({Mission} 1 : N {Evidence}) was implemented by adding {MissionID} as an FK to the {EVIDENCE} relation.

4. Mapping of M:N Relationship Types (Relationships with Attributes)

- Rule (M:N): For all Many-to-Many relationships ({AgentAssignment}, {CriminalInvolvement}), we created a new junction relation

- **Primary Key:** The PK for the junction relation is the composite key formed by the Primary Keys of all participating entities.
- **Example:** The {CriminalInvolvement} Primary Key became {CriminalID, MissionID}

5. Mapping of Multi-valued Attributes

- **Rule:** For attributes explicitly defined as multi-valued ({SkillSet} for {Criminal}, {Specialties} for {Agent}), we created a new junction table.
- **Example:** The {SkillSet} attribute was moved out to form CriminalSkillset{CriminalID, Skill}.

6. Mapping of Subclass Hierarchy (Generalization)

- **Rule:** We implemented the {Criminal} superclass and its Leader/Operative subclasses using Superclass and Subclasses.
- **Implementation:** We retained the {CRIMINAL} superclass table. We then created separate relations for {LEADER} and {OPERATIVE}, where the PK of each subclass is the {CriminalID} (which also acts as an FK back to the {CRIMINAL} superclass).

7. Mapping of Derived Attributes

- **Rule:** We omitted derived attributes ({DurationDays} in {Mission}, {ExperienceYears} in {Agent}) from the schema entirely, as they introduce data redundancy and update anomalies. These values are computed at query time.

2 - Relational Model To 1NF

No changes required because all entities were already in atomic form .

3 - 1NF To 2NF

No changes because no functional dependencies were found .

4 - 2NF To 3NF

No changes because no transitive dependencies were found .

RELATIONAL MODEL

Agent							
FirstName	LastName	Rank	<u>AgentID</u>	Country	HireDate	Email	Phone

Intelshare		
<u>CriminalID</u>	<u>ReceiverAgentID</u>	<u>SenderAgentID</u>

Mission						
<u>MissionID</u>	MissionName	Objective	StartDate	EndDate	Status	Region

Criminal					
<u>CriminalID</u>	Name	Alias	Nationality	Status	Member_Of

CriminalSkillset	
Skill	<u>CriminalID</u>

Leader		
<u>CriminalID</u>	CommandRegion	Leadership_Tier

Operative		
<u>CriminalID</u>	Specialty	HandlerID

Artifact					
<u>ArtifactID</u>	ArtifactName	OriginCountry	HistoricalValue	CurrentStatus	MissionID

Location						
<u>LocationID</u>	City	Country	Region	Latitude	Longitude	RiskLevel

Specialties	
Speciality_Name	<u>Speciality_ID</u>

ArtifactMovement						
HandledBy_AgentID	MoveReason	<u>MovementNo</u>	MovedOn	FromLocationID	ToLocationID	<u>ArtifactID</u>

Organization			
OrgName	Type	HQ_LocationID	<u>OrgID</u>

Evidence					
<u>MissionID</u>	ChainOfCustody	Type	Status	<u>EvidenceNo</u>	CollectedBy_AgentID

MissionTask				
<u>MissionID</u>	Description	<u>TaskNo</u>	AssignedAgentID	Status

AgentAssignment			
AssignedOn	Role	<u>MissionID</u>	<u>AgentID</u>

CriminalInvolvement				
Reward	Arrested	Role	<u>MissionID</u>	<u>CriminalID</u>

Tracks	
<u>CriminalID</u>	<u>AgentID</u>

Operation_logs			
<u>AgentID</u>	<u>ArtifactID</u>	<u>LocationID</u>	<u>MissionID</u>

AgentSkillset	
<u>AgentID</u>	<u>SpecialityID</u>

1NF

Agent							
FirstName	LastName	Rank	<u>AgentID</u>	Country	HireDate	Email	Phone

Intelshare		
<u>CriminalID</u>	<u>RecieverAgentID</u>	<u>SenderAgentID</u>

Mission						
<u>MissionID</u>	MissionName	Objective	StartDate	EndDate	Status	Region

Criminal					
<u>CriminalID</u>	Name	Alias	Nationality	Status	Member_Of

CriminalSkillset	
Skill	<u>CriminalID</u>

Leader		
<u>CriminalID</u>	CommandRegion	Leadership_Tier

Operative		
<u>CriminalID</u>	Specialty	HandlerID

Artifact					
<u>ArtifactID</u>	ArtifactName	OriginCountry	HistoricalValue	CurrentStatus	MissionID

Location						
<u>LocationID</u>	City	Country	Region	Latitude	Longitude	RiskLevel

Specialties	
Speciality_Name	<u>Speciality_ID</u>

ArtifactMovement						
HandledBy_AgentID	MoveReason	<u>MovementNo</u>	MovedOn	FromLocationID	ToLocationID	<u>ArtifactID</u>

Organization			
OrgName	Type	HQ_LocationID	<u>OrgID</u>

Evidence					
<u>MissionID</u>	ChainOfCustody	Type	Status	<u>EvidenceNo</u>	CollectedBy_AgentID

MissionTask				
<u>MissionID</u>	Description	<u>TaskNo</u>	AssignedAgentID	Status

AgentAssignment			
AssignedOn	Role	<u>MissionID</u>	<u>AgentID</u>

CriminalInvolvement				
Reward	Arrested	Role	<u>MissionID</u>	<u>CriminalID</u>

Tracks	
<u>CriminalID</u>	<u>AgentID</u>

Operation_logs			
<u>AgentID</u>	<u>ArtifactID</u>	<u>LocationID</u>	<u>MissionID</u>

AgentSkillset	
<u>AgentID</u>	<u>SpecialtyID</u>

2NF

Agent							
FirstName	LastName	Rank	<u>AgentID</u>	Country	HireDate	Email	Phone

Intelshare		
<u>CriminalID</u>	<u>ReceiverAgentID</u>	<u>SenderAgentID</u>

Mission						
<u>MissionID</u>	MissionName	Objective	StartDate	EndDate	Status	Region

Criminal					
<u>CriminalID</u>	Name	Alias	Nationality	Status	Member_Of

CriminalSkillset	
<u>Skill</u>	<u>CriminalID</u>

Leader		
<u>CriminalID</u>	CommandRegion	Leadership_Tier

Operative		
<u>CriminalID</u>	Specialty	HandlerID

Artifact					
<u>ArtifactID</u>	ArtifactName	OriginCountry	HistoricalValue	CurrentStatus	MissionID

Location						
<u>LocationID</u>	City	Country	Region	Latitude	Longitude	RiskLevel

Specialties	
<u>Speciality_Name</u>	<u>Speciality_ID</u>

ArtifactMovement						
HandledBy_AgentID	MoveReason	<u>MovementNo</u>	MovedOn	FromLocationID	ToLocationID	<u>ArtifactID</u>

Organization			
OrgName	Type	HQ_LocationID	<u>OrgID</u>

Evidence					
<u>MissionID</u>	ChainOfCustody	Type	Status	<u>EvidenceNo</u>	CollectedBy_AgentID

MissionTask				
<u>MissionID</u>	Description	<u>TaskNo</u>	AssignedAgentID	Status

AgentAssignment			
AssignedOn	Role	<u>MissionID</u>	<u>AgentID</u>

CriminalInvolvement				
Reward	Arrested	Role	<u>MissionID</u>	<u>CriminalID</u>

Tracks	
<u>CriminalID</u>	<u>AgentID</u>

Operation_logs			
<u>AgentID</u>	<u>ArtifactID</u>	<u>LocationID</u>	<u>MissionID</u>

AgentSkillset	
<u>AgentID</u>	<u>SpecialtyID</u>

3NF

Agent							
FirstName	LastName	Rank	<u>AgentID</u>	Country	HireDate	Email	Phone

Intelshare		
<u>CriminalID</u>	<u>RecieverAgentID</u>	<u>SenderAgentID</u>

Mission						
<u>MissionID</u>	MissionName	Objective	StartDate	EndDate	Status	Region

Criminal					
<u>CriminalID</u>	Name	Alias	Nationality	Status	Member_Of

CriminalSkillset	
Skill	<u>CriminalID</u>

Leader		
<u>CriminalID</u>	CommandRegion	Leadership_Tier

Operative		
<u>CriminalID</u>	Specialty	HandlerID

Artifact					
<u>ArtifactID</u>	ArtifactName	OriginCountry	HistoricalValue	CurrentStatus	MissionID

Location						
<u>LocationID</u>	City	Country	Region	Latitude	Longitude	RiskLevel

Specialties	
Speciality_Name	<u>Speciality_ID</u>

ArtifactMovement						
HandledBy_AgentID	MoveReason	<u>MovementNo</u>	MovedOn	FromLocationID	ToLocationID	<u>ArtifactID</u>

Organization			
OrgName	Type	HQ_LocationID	<u>OrgID</u>

Evidence					
<u>MissionID</u>	ChainOfCustody	Type	Status	<u>EvidenceNo</u>	CollectedBy_AgentID

MissionTask				
<u>MissionID</u>	Description	<u>TaskNo</u>	AssignedAgentID	Status

AgentAssignment			
AssignedOn	Role	<u>MissionID</u>	<u>AgentID</u>

CriminalInvolvement				
Reward	Arrested	Role	<u>MissionID</u>	<u>CriminalID</u>

Tracks	
<u>CriminalID</u>	<u>AgentID</u>

Operation_logs			
<u>AgentID</u>	<u>ArtifactID</u>	<u>LocationID</u>	<u>MissionID</u>

AgentSkillset	
<u>AgentID</u>	<u>SpecialtyID</u>